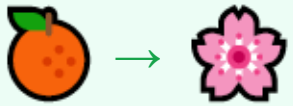
Metabolism drives perceptual transitions — Logic rules make them queryable, auditable, and falsifiable

Biological Process

ripening · fermentation · decay



enzymatic cascades transform molecules

EC 4.2.3 (lyase)

causes

Molecular Space

KEGG metabolic network

Limonene $C_{10}H_{16}$ — **4.2.3** — **Linalool** $C_{10}H_{18}O$

structure changes → volatiles change

→ **what reaches the nose changes**

perceived

Human Perception

TGSC labels = what the brain smells

citrus → **floral**

"it smelled citrusy, now it smells floral"

perceptual transition = brain's experience of metabolism

Framework: Cartesian Odor Event (S , E , T) + Reasoning Engine

Binds mechanism to perception · Supports deduction, abduction, consistency checking with proof traces

S = Source

"smells citrus"

→

E = Enzyme Sequence

[EC 4.2.3 → EC 1.1.1]

metabolic operator

→

T = Target

"smells floral"

Reasoning Queries

Q1 Forward: (S,E)→{T}

Q2 Backward: (S,T)→{E}

Q3 Consistency check

mine rules

Odor space is rule-governed: directional, boundary-limited, and mechanistically auditable

Hard Barriers

8,072 exclusion rules @ conf=1.0

categorical blocks under specific EC sequences

Order Sensitivity

[A→B] ≠ [B→A] — non-commutative

same enzymes, different order → different odor

Enzyme-Class Incompatibility

351 mutual-exclusion rules

some EC pairs never co-occur in viable pathways

Directional Gates

733 conditional-necessary, 566 asymmetric

breaks reversibility assumption

Sequence Dominance

fixed EC sequence → one target

funnels multiple sources to single outcome

Enzyme-Set Sufficiency

specific EC combinations → target odor

deterministic production rules